

Composite Solids, Density and Design

Answer Key

High School Geometry

Quick Answers

- | | |
|--------------------------|--|
| 1. 351.86 m^3 | 7. 2.12 m |
| 2. 0.42 m^3 | 8. 565.49 cm^2 |
| 3. 87.96 m^2 | 9. 498.11 g |
| 4. 1800 g | 10. (a) required height = 3.06 m , (b) mass of |
| 5. 282.74 m^3 | steel = 1217.43 kg , — |
| 6. 613.64 cm^3 | |
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What is verified

11 of 12 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-GMD.A.3 – problems 1, 2, 3, 5, 8
HSG-MG.A.2 – problems 4, 6, 9
HSG-MG.A.3 – problems 7, 10
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

- 2. *If they got 0.48: forgot to subtract the drilled hole*
 - 3. *If they got 100.53: included a second circle for the open top*
 - 5. *If they got 339.29: used the whole-sphere formula instead of a hemisphere*
 - 6. *If they got 47520: multiplied mass by density instead of dividing*
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1. Cylindrical silo, radius 4 m , height 7 m . Find the volume.

Solution: $V = \pi r^2 h = \pi(4)^2(7) = 112\pi = 351.8584\dots$, so

$V \approx 351.86 \text{ m}^3$

2. Block $1.2 \times 0.8 \times 0.5 \text{ m}$ with a cylindrical hole of radius 0.2 m drilled through the 0.5 m height.

Solution: Volume of concrete = block – hole. The hole is a cylinder whose height is the block's 0.5 m.

$$\begin{aligned} V &= (1.2)(0.8)(0.5) - \pi(0.2)^2(0.5) \\ &= 0.48 - 0.0628 \dots \end{aligned}$$

So

$V \approx 0.42 \text{ m}^3$

- 3.** Open-topped cylinder, radius 2 m, height 6 m. Find the sheet metal for wall + base.

Solution: The wall is a rectangle when unrolled, of area $2\pi rh$; the base is one circle, of area πr^2 . There is no top.

$$S = 2\pi(2)(6) + \pi(2)^2 = 24\pi + 4\pi = 28\pi = 87.9645 \dots$$

So

$S \approx 87.96 \text{ m}^2$

- 4.** Oak block, volume 2400 cm^3 , density 0.75 g/cm^3 . Find the mass.

Solution: Density is mass per unit volume, so mass = volume $\times$ density = 2400×0.75 .

1800 g

- 5.** Silo: cylinder radius 3 m, height 8 m, topped by a hemisphere of radius 3 m.

Solution: Add the two pieces. A hemisphere is half a sphere, so its volume is $\frac{1}{2} \cdot \frac{4}{3}\pi r^3 = \frac{2}{3}\pi r^3$.

$$\begin{aligned} V &= \pi(3)^2(8) + \frac{2}{3}\pi(3)^3 \\ &= 72\pi + 18\pi = 90\pi = 282.7433 \dots \end{aligned}$$

So

$V \approx 282.74 \text{ m}^3$

- 6.** Bronze statuette, mass 5400 g, density 8.8 g/cm^3 . Find the volume.

Solution: Rearranging density = mass $\div$ volume gives volume = mass $\div$ density.

$$V = \frac{5400}{8.8} = 613.6363 \dots$$

So

$V \approx 613.64 \text{ cm}^3$

- 7.** Cylindrical tank holding 15 m^3 with radius fixed at 1.5 m. Find the height.

Solution: The capacity is the constraint, so solve $\pi r^2 h = 15$ for h .

$$h = \frac{15}{\pi(1.5)^2} = \frac{15}{2.25\pi} = 2.1220 \dots$$

So

$$h \approx 2.12 \text{ m}$$

8. Hopper: cone radius 5 cm, slant 13 cm, on a closed-bottom cylinder radius 5 cm, height 9 cm.

Solution: Three surfaces are exposed — the cone's curved surface, the cylinder's curved wall, and the cylinder's base. The disc where the two solids meet is inside the solid and is not painted.

$$\begin{aligned} S &= \pi r \ell + 2\pi r h + \pi r^2 \\ &= \pi(5)(13) + 2\pi(5)(9) + \pi(5)^2 \\ &= 65\pi + 90\pi + 25\pi = 180\pi = 565.4867 \dots \end{aligned}$$

So

$$S \approx 565.49 \text{ cm}^2$$

9. Glass cube of edge 6 cm with a hemispherical hollow of radius 2 cm; density 2.5 g/cm³.

Solution: Find the glass volume first, then convert to mass with the density.

$$\begin{aligned} V &= 6^3 - \frac{2}{3}\pi(2)^3 = 216 - 16.7551 \dots = 199.2448 \dots \\ m &= 2.5 \times 199.2448 \dots = 498.1121 \dots \end{aligned}$$

So

$$m \approx 498.11 \text{ g}$$

10. Grain bin, radius 2.5 m, capacity 60 m³, steel 18 kg per m² on wall and base only.

Solution (a): The capacity fixes the height:

$$h = \frac{60}{\pi(2.5)^2} = \frac{60}{6.25\pi} = 3.0557 \dots$$

So

$$h \approx 3.06 \text{ m}$$

Solution (b): Steel area = curved wall + base. Keeping h exact avoids rounding error:

$$\begin{aligned} \text{wall} &= 2\pi(2.5)h = 2\pi(2.5) \cdot \frac{60}{6.25\pi} = 48 \text{ exactly} \\ \text{base} &= \pi(2.5)^2 = 19.6349 \dots \\ \text{mass} &= 18(48 + 19.6349 \dots) = 1217.4292 \dots \end{aligned}$$

So

$$m \approx 1217.43 \text{ kg}$$

Solution (c) — instructor-judged. A model response: the capacity is fixed, so the height is not free — it is $60/(\pi r^2)$, and doubling the radius divides the height by four. The curved wall is $2\pi r h$, and since r doubles while h is quartered the wall area actually *halves*; but the base is πr^2 , which is multiplied by four, from about 19.63 to about 78.54 m². Working it through, the wall drops from 48 to 24 m² while the base rises from about 19.63 to about 78.54, so the total area rises from about 67.63 to about 102.54 m² and the steel goes *up*, not down.

Full credit: says the height is forced by the fixed capacity and separates the wall's behaviour from the base's. *Half credit:* says only that the base grows with r^2 . *No credit:* treats the height as unchanged.